

Multiple Choice (10 marks)

1. Evaluate the series $S = \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \cdots$
 - (a) 0.5
 - (b) 2.5
 - (c) 1.25
 - (d) 1.5
 - (e) 4
2. A chemist wants to produce chlorine from molten NaCl. How many grams could be produced if he uses a current of one amp for 5.00 minutes?
 - (a) 0.031 g
 - (b) 5.55 g
 - (c) 0.021 g
 - (d) 0.511 g
 - (e) 0.111 g
3. 2.000 picogram(pg) of ^{33}P decays by β^- emission to ^{33}S in 75.9 days. Find the half-life of ^{33}P .
 4. 37.95 days
 5. 50.6 days
 6. 8 days
 7. 5.6 days
 8. 60.4 days

The photoelectron spectrum of carbon has three equally sized peaks. What peak is at the lowest energy?

- (a) The 2 p peak has the lowest energy.
- (b) The 3 d peak has the lowest energy.
- (c) The 2 d peak has the lowest energy.
- (d) The 1 d peak has the lowest energy.
- (e) The 1 p peak has the lowest energy.

Calculate the voltage of the cell $\text{Fe}; \text{Fe}^{+2} || \text{H}^+; \text{H}_2$ if the iron half cell is at standard conditions but the H^+ ion concentration is 0.1 M.

0.50 volts

0.76 volts

1.2 volts

0.33 volts

.85 volt

True / False (10 marks)

Answer True or False

6. True or False: Thionyl chloride (SOCl_2) has a tetrahedral molecular geometry due to its four bonded atoms around the sulfur atom.
7. True or False: The Q-factor for an X-band EPR cavity with a resonator bandwidth of 1.58 MHz is $Q = 5012$.
8. True or False: The value of $(0.0081)^{3/4}$ is equal to $(1000) / (81)$.
9. True or False: To prepare a 50 ml solution of pH 1.25 HCl, one must add 49.76 ml of H_2O to 0.24 ml of concentrated HCl.
10. True or False: The molarity of the sodium hydroxide solution is 0.0641 M after neutralization with hydrochloric acid.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Calculate the pH of a solution formed by mixing 50.0 mL of 0.0025 M HBr with 50.0 mL of 0.0023 M KOH. Discuss the steps involved in your calculation and the underlying principles.
12. A particle of mass 9.1×10^{-28} g in a one-dimensional quantum box transitions from the $n = 5$ to $n = 2$ level, emitting a photon of frequency $6.0 \times 10^{14} \text{ s}^{-1}$. Calculate the length of the box and discuss the significance of this length in the context of quantum confinement.

End of Paper